

business solutions framework with the help of FOSS. If you have good expertise in the FOSS terrain, you can develop a distributed, highly customised architecture very easily. You can make the best of your available resources due to the adaptable nature of OSS. And this is what we did at Naaptol. On a day-to-day basis, use of OSS in different departments of the business boosts your solution analysis power." Patil states that the team always keeps track of the current market trends with reference to proprietary software and tries to find an alternative open source tool.

## A comprehensive list of FOSS tools used at Naaptol

### Application platform

| Software/Tool/<br>Framework  | Short description/Purpose                                                                                                                                                               |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Apache Http server           | Web server                                                                                                                                                                              |
| MySQL                        | Database                                                                                                                                                                                |
| HAProxy                      | Load balancer                                                                                                                                                                           |
| JBoss                        | Application server                                                                                                                                                                      |
| Apache SOLR                  | Enterprise search engine<br>In addition, the firm is also in process of evaluation and migration to SOLR Cloud, which is an enterprise-class distributed search platform on top of SOLR |
| JSF                          | Framework for building component-based user interfaces for Web applications                                                                                                             |
| Hibernate                    | Framework for object-relational mapping (ORM)                                                                                                                                           |
| Velocity                     | Template engine                                                                                                                                                                         |
| Ehcache / OSCache / Memcache | Caching frameworks                                                                                                                                                                      |
| Apache Axis                  | Web service framework                                                                                                                                                                   |
| Phonegap / Apache Cordova    | Mobile development framework                                                                                                                                                            |
| Apache Mobile Filter (AMF)   | Mobile device detection filter                                                                                                                                                          |
| Mod_security                 | Web application firewall                                                                                                                                                                |

### Infrastructure

| Software/Tool/<br>Framework | Short description/Purpose                  |
|-----------------------------|--------------------------------------------|
| CentOS                      | Enterprise-class Linux OS for servers      |
| Nagios                      | Monitoring tool for the network and server |
| OSTicket                    | Support ticket system                      |

### Development environment

| Software/Tool/<br>Framework | Short description/Purpose                |
|-----------------------------|------------------------------------------|
| Ubuntu                      | Desktop environment                      |
| CVS                         | Versioning system                        |
| Eclipse                     | Integrated development environment (IDE) |
| Apache Ant                  | Build management tool                    |

### QA/Testing environment

| Software/Tool/<br>Framework | Short description/Purpose                                 |
|-----------------------------|-----------------------------------------------------------|
| Bugzilla                    | Bug-tracking system                                       |
| selenium                    | Function testing                                          |
| JMETER                      | To load test functional behaviour and measure performance |
| EMMA/JaCoCo                 | Code coverage tool                                        |

### Performance/Benchmarking/Debugging/Analysis

| Software/Tool/Framework                                                  | Short description/Purpose                                     |
|--------------------------------------------------------------------------|---------------------------------------------------------------|
| Maatkit                                                                  | For MySQL database analysis, rectification and administration |
| SysBench                                                                 | System performance benchmarking                               |
| Java Interactive Profiler (JIP) / JBoss Profiler / Memory Analyzer (MAT) | A profiling tool                                              |

### Major challenges faced while deploying FOSS

Industry experts believe that when it comes to adopting FOSS in companies, one does face a number of challenges. When we put this question to Patil, he did agree, saying, "You might face difficulties in the adoption of open source software if you have not done the right evaluation or perhaps conducted a survey; or you are not clear about your own requirements before the actual implementation of FOSS tools. We didn't have any major challenges, as such. When a software's structure is changed in a new release, there are chances that you face difficulties in the upgradation process. At Naaptol, we always keep it in mind that we do all the testing of hardware and application platform portability for greater scalability. It also depends upon the implementation methodology. So, while deploying an open source solution in your organisation, you need to fix such issues on a constant basis while using FOSS."

### Advantages that FOSS brings to companies

From his own experience with open source technology, Patil talks about the significant benefits that FOSS can render to companies looking to go the open source way. Here are some pointers from him, as told to us.